

Continued Disproof Sentence towards Collatz Conjecture

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ABSTRACT

We are to address the recent critical insight on our key idea towards asymptotic disproof of Collatz conjecture by estimating of the algorithmic step-wise approach of computing numbers in this sequence, which finally converge to one.

Keywords: Collatz conjecture, disproof, statement, approach.

INTRODUCTION

The Collatz conjecture is a long-standing and well defined question in mathematics [1], within the proposed solution till now [2-4].

It generally states if there is an approximation of the relation to the finite one number, also there was an approach to provide a constructive invalidation of our obtained results [5], however, we are to give the definite and concise answer for all the undefined questions in our proposed solution towards Collatz conjecture.

BIG O-NOTATION

As we define $O(f(n))$ to be the number of steps in recurrent relation to be equal or less towards the number n itself, obviously, our key idea lays in proving that this condition holds true for any positive integer number and fails otherwise, thus giving inconsistent fact if there is a loop or, in other words, infinite number of steps, thus giving the fact that the recurrent relation never converges to the number one.

We have to note the correct interpretation of $O(f(n))$ as a number of steps in iteration, thus we have:

$$O(f(n)) \rightarrow \infty : O(f(n)) = |A|, A = \{n : n \in f(n)\}.$$

Where in the above in correct interpretation of our statement, $O(f(n))$ is the number of steps in recurrent approximation of Collatz sequence and A is the set of all numbers in this sequence for the any natural number n : obviously there are two basic scenarios – first is when there's a loop and second is when the sequence isn't converging completely and, thus, isn't limited.

CONCLUSION

We have given a general notion towards the correct and probably more correct interpretation of our key idea towards the proof of Collatz conjecture using the algorithmic step-wise estimation of the Collatz function $f(n)$, which is better defined as $O(f(n))$.

Thus, the 'contra' argument by Mary Lafontaine and Gerard Cheong in [5] cannot be considered as final statement as we have extended our key idea towards the proof for better evaluation where the step-wise logic can be considered as the number of steps in Collatz sequence during approximation and not as the 'set of functions' in a classical and general understanding of O-notation, which can be adopted in the sentence of the more logical and definitive term – thus, any key idea is the right which is already proven by the fact that we are not to state that our proof isn't a 'claim', but an another 'deep-dive' into the computational logic and, thus, cannot be just 'overthrown' only for the reason that there's 'something incorrect or misinterpreted' – this would be our final word justified by the above explanation of our usage and interpretation of big O-

notation giving the truth if go beyond the barriers of classical interpretation of this big O-notation.

This is our final notion which gives that ‘obvious interpretations’ cannot be considered final and complete, since operational logic and O-notation generally is used to describe the algorithmic complexity and we, from there on, are going beyond that adopting it as a number of steps in Collatz sequence during each of the computation step which gives the analogy towards computational complexity, infinity and, surely, mathematical sense.

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